

Anitha Raj Bale

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SUMMARY

Senior Data Engineer with 5+ years of experience designing scalable data platforms and production pipelines across financial, healthcare, and AI-driven environments, progressing from core data engineering to intelligent data infrastructure. Combines Python, SQL, Spark, Databricks, Snowflake, AWS, and modern orchestration with ML-ready pipelines, feature engineering, MLflow, GenAI, LLM processing, RAG, and vector-based retrieval to build reliable data systems for enterprise AI applications.

TECHNICAL SKILLS

Programming & Querying: Python, SQL (Advanced Optimization, Window Functions), Java, Scala, Bash, JSON

Data Engineering & Processing: ETL/ELT, Data Modeling, Dimensional Modeling, Apache Spark (PySpark, Spark SQL, Structured Streaming), Apache Kafka (Kafka Connect, Schema Registry), Apache Airflow, Change Data Capture (CDC)

Cloud & Data Platforms: AWS (S3, EMR), Databricks (Delta Lake, Unity Catalog, Delta Live Tables, Databricks Workflows), Snowflake (Snowpark), Microsoft Fabric, Apache Iceberg, Parquet, Avro

AI/ML & Intelligent Data: Machine Learning Pipelines, Feature Engineering, ML-Ready Data Pipelines, MLflow, Model Training, Inference & Model Serving Pipelines, Generative AI Data Workflows, LLM Data Processing, LLM/API Integration, AI Application Integration, Embeddings, Vector Databases (Pinecone, pgvector), Vector Search, RAG (Retrieval-Augmented Generation) Pipelines

DevOps & Platform Engineering: Git, CI/CD (GitHub Actions, Jenkins), Docker (Docker Compose, Multi-stage Builds), Kubernetes (Pods, Deployments), Terraform (IaC), Linux, REST APIs

Data Quality, Databases & Analytics: Data Quality, Data Governance, Data Lineage, Data Observability (Great Expectations), Query Optimization, Performance Tuning, Partitioning, Monitoring, PostgreSQL, MySQL, Tableau, Power BI

PROFESSIONAL EXPERIENCE

Senior AI/ML Data Engineer, *Pinecone*

Jun 2025 - Present | Remote, USA

- Architected Machine Learning Pipelines for recommendation services, combining Feature Engineering with MLflow tracking and ML-ready datasets across 35 feature groups for production model preparation workflows.
- Operationalized model training, inference and model serving pipelines for intelligent search, improving deployment readiness through standardized validation, feature availability, and repeatable serving workflows for production applications.
- Launched Generative AI data workflows for support-document intelligence, transforming LLM data processing into inference pipelines that handled 1.6 million documents monthly with systematized enrichment controls.
- Connected Pinecone and pgvector vector databases, generating embeddings and implementing vector search that improved relevant-context retrieval by 43% across indexed enterprise knowledge collections and semantic workloads.
- Established RAG (Retrieval-Augmented Generation) pipelines with LLM/API integration, exposing REST APIs that connected enterprise retrieval with production AI assistants while supporting secure model-serving workflows.
- Containerized AI applications using Docker (Docker Compose, Multi-stage Builds), integrating Git and GitHub Actions while strengthening Data Observability, AI application integration, and production monitoring capabilities.

Senior Data Engineer, *GE HealthCare*

Aug 2022 - Nov 2023 | Remote, India

- Led Databricks modernization for patient-utilization analytics, applying Delta Lake and Medallion Architecture to organize healthcare records across governed Bronze, Silver, and Gold layers for clinical analytics.
- Deployed Unity Catalog with Delta Live Tables, strengthening Data Governance while increasing trusted dataset coverage from 82% to 97% through mechanized controls and standardized access.
- Designed Snowflake warehouse structures with Snowpark transformations, improving cohort-query response times by 38% while supporting scalable healthcare Data Warehouse reporting across multiple clinical programs enterprise-wide.
- Computerized Databricks Workflows with Jenkins releases, reducing deployment errors through standardized validation, promotion, rollback, and release procedures across healthcare analytics services and production environments consistently.
- Provisioned Terraform infrastructure with Kubernetes Pods and Deployments, improving resource scalability while strengthening Linux-based recovery procedures for repeatable healthcare platform operations across production environments reliably.
- Integrated Apache Iceberg with Parquet and Avro datasets, establishing Great Expectations validation and Data Lineage monitoring while reducing schema-related incidents by 29% across shared clinical datasets.

Data Engineer, *IBM*

Sept 2019 - Jul 2022 | Remote, India

- Engineered Python ETL pipelines for banking transaction reconciliation, processing 12 million monthly records and improving settlement validation by 28% through standardized transformation and exception handling workflows.
- Developed SQL data models for credit-risk reporting, applying window functions and dimensional modeling to consolidate customer exposures while reducing recurring report preparation time by 35%.
- Built Java ingestion services for payment events, integrating JSON payloads with PostgreSQL sources and automated validation that improved downstream data consistency across daily financial reporting operations.
- Streamlined Apache Spark processing with PySpark and Spark SQL for portfolio analytics, reducing batch execution time by 24% through partitioning, caching, and targeted query optimization.
- Configured Apache Kafka pipelines for transaction-event streaming, establishing Schema Registry controls that improved message compatibility and supported reliable downstream fraud-monitoring workflows across high-volume financial systems.

- Orchestrated Apache Airflow workflows for regulatory-data preparation, coordinating validation steps and scheduled dependencies while improving pipeline completion reliability to 99.5% across recurring compliance reporting cycles.
- Implemented AWS S3 ingestion with EMR for financial datasets, applying Bash automation and CDC processing to shorten source-to-report latency by 47 minutes across daily operations.

PROJECTS

Retail Data Engineering Platform | Python, Kafka, Spark, Airflow, MySQL, Tableau

- Constructed a real-time retail data pipeline using Kafka and Spark Structured Streaming, transforming sales and inventory events through Bronze, Silver, and Gold processing layers.
- Arranged scheduled Airflow workflows to load curated MySQL datasets and deliver Tableau-ready reporting models supporting inventory visibility, sales analysis, and replenishment decision-making.

Microsoft Fabric Retail Sales Analytics | Microsoft Fabric, PySpark, SQL, Power BI

- Enhanced a Lakehouse-based retail analytics solution in Microsoft Fabric, applying PySpark transformations to cleanse, standardize, and organize sales data across Bronze, Silver, and Gold layers.
- Created SQL Analytics Endpoint queries and Power BI dashboards that converted curated sales datasets into interactive views of revenue trends, product performance, and regional activity.

Eventify – Master's Capstone Project

- Planned the relational data model and backend workflows for an event-management platform supporting event creation, ticket inventory, attendee registration, and transaction-related application records.
- Integrated REST APIs and authentication controls to automate user registration, event booking, and ticket-processing workflows while maintaining secure access to platform features and stored information.

IoT-Based Voice Controlled Smart Home

- Expanded a Raspberry Pi-based home automation system that translated Google Home voice commands into real-time control actions for connected household appliances and devices.
- Organized IFTTT-triggered communication workflows between voice services and IoT hardware, enabling remote device interaction and reliable execution of predefined smart-home automation commands.

CERTIFICATIONS

- Data Engineering Professional Certificate - Snowflake | Sep 2026 | ([Link](#))
- AWS Certified Data Engineer - Associate (DEA-C01) - AWS | ([Link](#))
- Data Engineering Foundation Certification - Informatica | Aug 2026 | ([Link](#))
- Databricks Fundamentals Accreditation - Databricks | ([Link](#))
- AWS Cloud Practitioner Essentials ([Link](#))

EDUCATION

Master of Science, Pace University Computer Science	Jan 2024 - Dec 2025 NY, USA
Bachelor of Technology, PVPSIT Information Technology	Jun 2015 - March 2019 AP, India